

JOB QUEST

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PROBLEM

Problem Statement

How might we help graduating students stay motivated and be successful while navigating their job search journey?

Research

Research

Conducted surveys, user interviews, literature review, and an expert interview to understand both the practical and emotional challenges graduating students face during the job-search process.

Used these findings to identify key pain points and inform the design of Job Quest's features and overall goals.

Literature Review

“Specific, clear job-search goals result in more effort and persistence, and a higher probability of performing well” (Hooft et al., 2020).

“Self-motivation was negatively correlated with time spent searching for employment” (Liu et al., 2014).

User Interviews & Surveys

Graduating Students and Early-Career Job Seekers

- Difficulty staying motivated during long job searches
- Frustration with rejection and lack of employer feedback
- Uncertainty about qualifications and competitiveness
- Desire for clearer progress tracking and structure
- Interest in tools that make the job search feel less overwhelming

Key Takeaways

- Job search challenges are both practical and emotional.
- Students need support with motivation, confidence, and persistence.
- Progress tracking and feedback are highly valued.
- Users want guidance that helps make the process feel more manageable.

Domain Expert Interviews

Ruyan Li – Peer Career Coach (UW Career & Internship Center).

Feedback is difficult for companies to provide due to:

- High application volume
- Candidates often being rejected because another candidate was stronger
- Concerns about discouraging future applicants

Many students have unrealistic expectations about entry-level tech jobs and hiring timelines.

Key Takeaways

- Lack of feedback is often a structural issue, not a reflection of applicant quality.
- New graduates need help setting realistic expectations for the job search.
- Career tools should focus on guidance and support, not just job listings.

Personas

JORDAN



Graduating Senior at UW
Computer Science
Aspiring Software Engineer

CHARACTERISTICS

- Has internship experience
- Has technical project experience
- Is frustrated that despite satisfying job listing requirements he is constantly facing rejection even for companies he really wants to work for.

GOALS

- Land a full-time SWE role at a company he is genuinely excited about
- Understand what is actually preventing him from converting applications into offers

PAIN POINTS

- Lacks actionable feedback to understand where his application is falling short
- Struggles to stay organized across a high volume of applications
- Internalizes rejection personally rather than attributing it to competitive market conditions
- Unsure whether to apply broadly or focus on fewer targeted applications
- Demotivated by the application process

BACKGROUND

Jordan is a recent computer science graduate with internship experience and multiple coding projects. Although he meets the qualifications for many entry-level software engineering roles, he frequently faces rejection without receiving feedback. He struggles to determine whether the issue is his resume, interview performance, application strategy, or the competitive job market. Over time, these repeated rejections have lowered his confidence and made it difficult to stay motivated and consistent in his job search.

MARA



Graduating senior at UW
Informatics
Aspiring UX designer

CHARACTERISTICS

- Doesn't have internship experience
- Doesn't know how to showcase skills
- Doesn't have existing network connections

GOALS

- Land her first UX job
- Learn how to showcase her skills successfully
- Learn how to communicate her experience confidently

PAIN POINTS

- Doesn't understand how ATS screening filters her out before human review
- Lacks access to informal industry knowledge her networked peers take for granted
- Struggles to stay consistent in applying amid repeated discouragement
- Unsure whether to keep applying now or invest more time building experience first

BACKGROUND

Mara discovered UX design late in college and has built experience through coursework and student organizations. Although she has a portfolio, she is unsure whether applicant tracking systems are filtering her resume before employers see it. Without a strong professional network, she struggles to understand what employers expect and how to effectively present her experiences. Repeated rejections and a lack of meaningful feedback have lowered her confidence, leaving her uncertain about whether she needs stronger qualifications or simply better application strategies.

SOLUTION

<https://alipson.me/projects/job-quest/>

[Video Link](#)

Problem To Solve

Sustain motivation & consistency while facing repeated rejection

Provide feedback to counter demotivation

Organize through high-volume application activity

Learn informal industry knowledge typically absorbed through networks

Our Solution: JOB QUEST

Web browser game & tool

Motivates job-seeking students to apply to jobs through a gamified experience

Provides a streamlined and optimized job application portal



Platform Justification

- **Sustained engagement:** gamification is the studied medium for tasks (Duolingo, Strava precedent).
- Self-regulation theory backs engagement-driven job-search activity (Hooft et al., 2020)
- **Externalize rejection:** break the personalization spiral; practice loop, not a fail state.
- **Visible progression:** “.. increased perceived progress was positively related to subsequent job search.” (da Motta Veiga & Turban, 2018)
- **Modality match:** between desktop game to desktop application work in browser
- **Versatile video game medium:** native affordances to address multifarious pain points

Key Features (Application)

Integrated application
resources must match or
exceed competing products

1. Centralized Information for Resume Personalization
2. Automating Repetitive Tasks
3. Integrated Logging & Application Tracking

Centralized Info Exhibit 1

Time elapsed & number of
applicants for fast filtering
of stronger listings

Integrated search of
relevant company and job
info needed for resume
catering

Role	Posted	Applicants	Match
Data Engineer, New York NEW	00:31:06	3 apps	91%
Software Engineering Campus Hire — NYC HOT	00:52:26	7 apps	88%
Junior Data Engineer — Series A Fintech, NYC LOW COMP	01:30:46	2 apps	83%
Data Engineer New Grad — Payments Analytics LOW COMP	02:29:06	4 apps	79%

MARKET INTELLIGENCE

0.05%

Predicted odds

0.1%

Accept rate

VERY HIGH

Competition

Aggregated from Reddit · Glassdoor · Levels.fyi · WSO

COMMUNITY INTEL

REDDIT

COMPANY SITE

Figma frontend interview: 1 LC medium, 1 UI component build from scratch, 1 system design for collaborative editing.

r/webdev · 441 upvotes · 1 week ago

GLASSDOOR

COMPANY SITE

Very product-focused engineering culture. Expected to have opinions on UX.

Glassdoor · ★★★★★ · 3 weeks ago

LEVELS.FYI

COMPANY SITE

IC3: \$160k-\$175k base + \$120k-\$150k RSU over 4yr.

Levels.fyi · 9 verified datapoints

RESUME BOOSTERS — WHAT GETS CALLBACKS AT THIS COMPANY

Performance

Figma's canvas is performance-critical. Any browser rendering optimization, canvas/WebGL experience, or profiling work stands out from the standard frontend applicant pool.

Collab

Figma explicitly values product-minded engineers. Show examples of working cross-functionally with designers or PMs.

Centralized Info Exhibit 2

Interaction with integrated ATS
(Applicant Tracking System)
simulator gives user direct
real-time feedback while keeping
full control of resume quality

Beats fully automated methods
such as AI apply by conserving
resume catering process but in
optimized form

Base Resume — Vladimir_Radostev_Base.pdf

BASE

40% match

CURRENT RESUME

Analysis auto-loaded for this listing.
Upload your PDF to preview it.

Preview base PDF

Base resume loaded

Click to upload your PDF and
preview it here

Upload Tailored Resume

Re-scan with your company-specific
version to see updated analysis

ATS EXTRACTED FIELDS

What the ATS bot reads from your file

NAME	Vladimir Radostev ✓
EMAIL	viadrad@uw.edu ✓
DEGREE	BS CS ✓
TYPESCRIPT	Not found X
REACT	Not found X
JAVASCRIPT	Not found X
WEBGL	Not found X
CANVAS	Not found X
FRONTEND	No experience X
D3.js	Mentioned in bullets ✓
PERFORMANCE	Latency reduction ✓

ATS SCORE VS LISTING

48%
ATS fit

AI FIT ANALYSIS

Evaluating Vladimir_Radostev_Base.pdf vs. Figma

✓ D3 visualization work is relevant

You built an interactive D3 dashboard for RL research. Figma's canvas team does heavy visualization work — D3 shows you can navigate browser-rendered graphics.

✓ Performance work on record

45% latency reduction + 50% training time reduction shows performance-minded engineering, which Figma values in canvas work.

X TypeScript and React absent — critical

Figma's frontend is TypeScript + React. These are required keywords. Neither appears in the base resume. This is the biggest mismatch.

X No frontend experience visible

The base resume has no web application or frontend project history. Figma requires production web shipping experience.

X WebGL/Canvas entirely missing

Figma's canvas is WebGL-powered. Any GPU rendering, canvas API, or graphics work is a differentiator here.

— This is a weak match overall

ATS EXTRACTED FIELDS

What the ATS bot reads from your file

NAME	Vladimir Radostev ✓	
EMAIL	viadrad@uw.edu ✓	
DEGREE	BS CS ✓	
PYTHON	Skills + bullets ✓	
SQL	Skills + depth ✓	
PIPELINES	Found in bullets ✓	
OWNERSHIP	End-to-end framing ✓	↑ improved
KAFKA	Not found X	
FINTECH	Financial modeling ✓	↑ improved
STARTUP EXP	AG2.AI startup context ✓	↑ improved
SQL DEPTH	Window functions, CTEs ✓	↑ improved

ATS SCORE VS LISTING

91%
ATS fit

AI FIT ANALYSIS

Evaluating Vladimir_Radostev_Cascade.pdf vs. Cascade Data

✓ Startup ownership framing now leads

AG2.AI startup experience frames the engineering context correctly: small team, direct ownership, no safety net — exactly what Cascade evaluates for.

✓ Fintech and financial modeling signals added

Murphy's Pub stochastic model — Markov processes, Monte Carlo simulation, POS data analysis — reads as quant trading desk data work in this context.

✓ SQL depth now demonstrated

Window functions, CTEs, and query optimization now surface in the project description rather than just appearing in the skills list.

✓ End-to-end delivery established

Multiple bullets now show zero-to-production delivery: problem → pipeline → result. Series A companies hire for this.

X Kafka still absent

Cascade's job description mentions streaming frameworks. If you have any Kafka exposure, mention it. Otherwise it remains a minor gap.

Task Automation

Headless automated applications allows users to apply to more jobs without waiting for automations to finish (not offered by Simplify)

Ready to submit

ATS Score **78%**
Company **Meta**
EEO pre-fill **✓ Ready**
Resume file **✓ Loaded**

Agent auto-fills all standard form fields. Your resume submits exactly as uploaded — no edits made by Job Quest.

Submit Application

EEO Pre-fill Setup

Enter once. Job Quest auto-fills these fields on every application — you never retype them. Valid for 24 hours per session.

GENDER

Prefer not to say

RACE / ETHNICITY

Prefer not to say

VETERAN STATUS

Prefer not to say

DISABILITY STATUS

Prefer not to say

WORK AUTHORIZATION

US Citizen

LINKEDIN URL

linkedin.com/in/yourprofile

Stored for 24 hours. Update any time in settings.

Company	Role	Date	Status	Notes
Meta	Data Engineer, NY	Today	SENDING...	▶ Agent active
Airbnb	Platform Engineer	May 9	PENDING	—
Coinbase	Backend SWE	May 7	REJECTED	Auto-detected

Logging

Automatic application logging & status updates via email integration
Simplifies tracking applications more than any other service & further
establishes all-in-one functionality

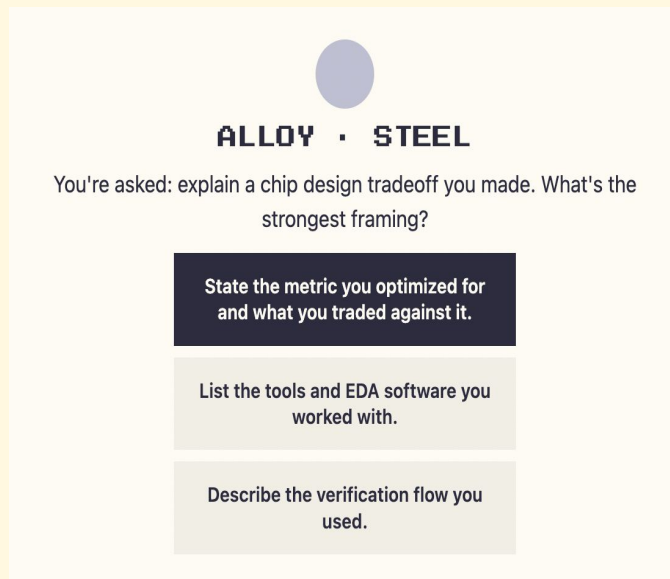
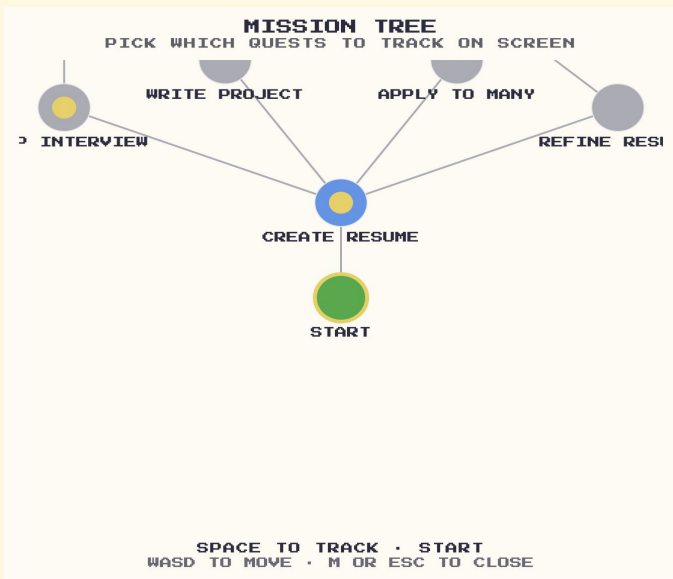
Application Log				
Updated automatically from your connected email. Rejections, OA invites, and interview requests are auto-detected.				
Company	Role	Date	Status	Notes
Meta	Data Engineer, NY	Today	SENDING...	▶ Agent active
Airbnb	Platform Engineer	May 9	PENDING	—
Coinbase	Backend SWE	May 7	REJECTED	Auto-detected
Stripe	SWE Infrastructure	May 6	OA RECEIVED	Due 18hrs
Figma	Frontend Engineer	May 5	INTERVIEW	May 14
Microsoft	Data Engineer	May 3	REJECTED	Auto-detected
Two Sigma	SWE Campus Hire	May 2	PENDING	—

Key Features (Gamification)

Modern popular apps use
attention maximization
methods

1. Quests, Achievements, and Questions.
2. Shareable Metrics (Response rate and # applications)

Quests, Achievements & Questions



Job Quest uses quests, achievements, and question encounters to make career preparation feel more engaging. Quests guide users through job-search tasks, achievements reward progress, and question encounters provide low-pressure opportunities to practice career-related skills. Together, these features help break the job search into smaller, achievable goals while keeping users motivated.

Shareable Metrics

Shareable Metrics allow users to track their job-search progress through statistics such as applications submitted, response rate, interviews received, and ongoing opportunities. These metrics help users visualize their progress, stay motivated, and celebrate milestones throughout the job-search process.

47

Apps this week

+6 today

8.5%

Response rate

+1.2% vs avg

4

Interviews

2 scheduled

3

OAs pending

due this week

Shareable Metrics

Friends

Application activity among people you know. Everyone's in the same market.



Alex K.

CS '26 · targeting SWE

LVL 9

61 apps · 11.2% response



You

CS '26 · targeting SWE / Data

LVL 7

47 apps · 8.5% response



Priya M.

Informatics '26 · targeting UX

LVL 6

38 apps · 7.9% response



Jordan W.

CS '26 · targeting SWE

LVL 6

34 apps · 6.1% response



Sam T.

Math '26 · targeting quant

LVL 5

22 apps · 9.1% response



EVALUATION

Usability Testing

Does the game experience feel
intuitive, engaging, and motivating,
while still being a practical job-search tool?

Testing Protocols

- Participant panel included 6 graduating seniors who are actively in job search process
- Latest 3 participants were moderated in-person or remotely via video call and screen-sharing
- Testing performed in five stages:
 1. Pre-gameplay questions on gaming familiarity and job-search confidence
 2. Directed and observed gameplay tasks and task-specific reflection questions
 3. Application Arena observed gameplay, tasks, and evaluation questions
 4. Overall Likert 1-5 satisfaction scale questions
 5. Summary reflection

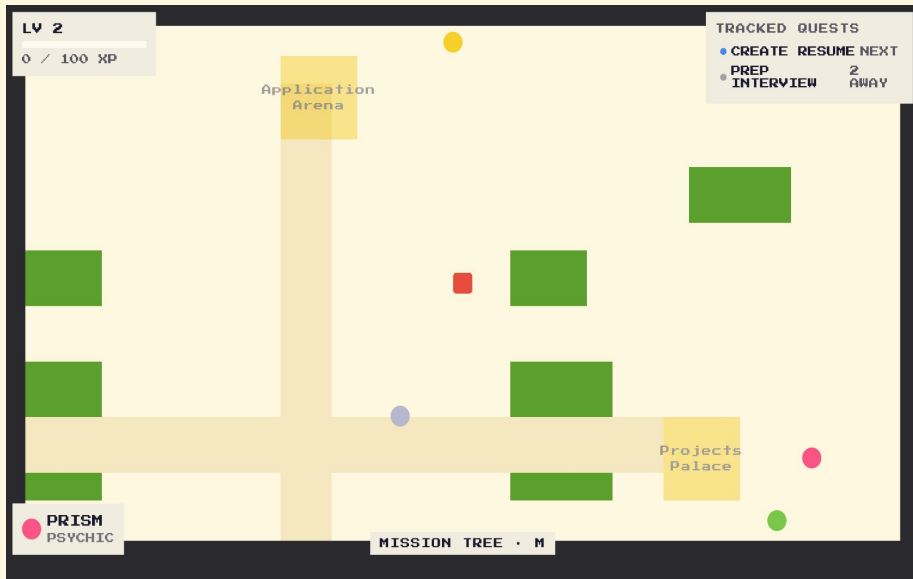
Testing Takeaways

- Participants reflected target audience: 4 aligned with Jordan, 2 with Mara
- Mixed in experience with video games
- All framed gamification positively
- Engagement when exploring is strongest (4.50/5)
- Moment-to-moment guidance is weakest (3.50/5)
- Substance gaps in Resume & Interview rooms: desired depth, not redesign
- Mission tree comprehension was recovered by last 3 testers post-update
- Player spawn placement is the most-cited unprompted issue
- Creatures were bimodal: a standout feature when read, invisible when not; only fully recognized by half of participants

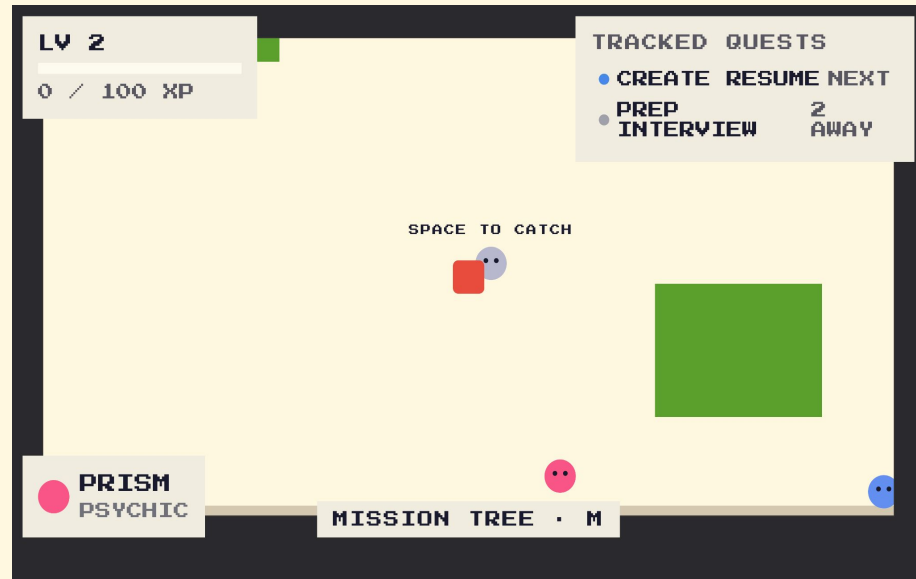
Design Updates

Creature Catch

Before



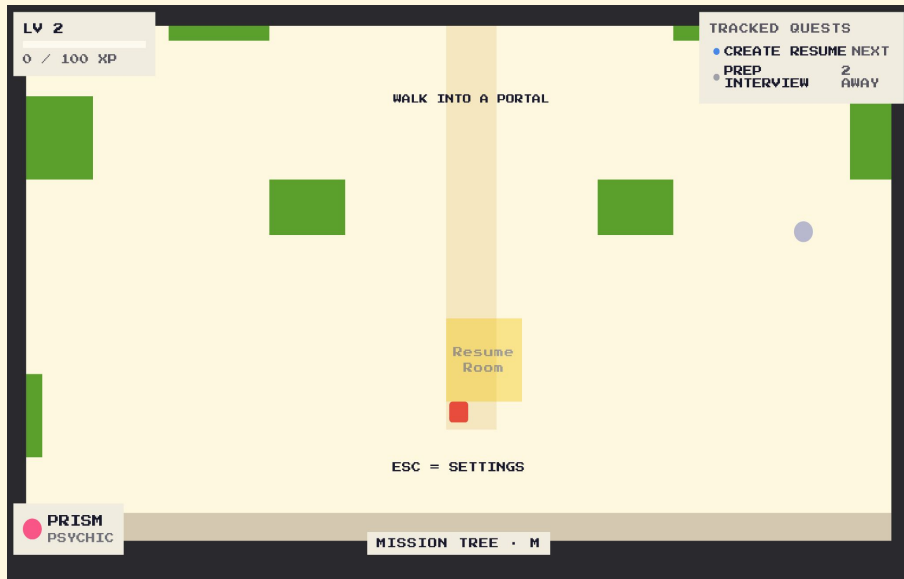
After



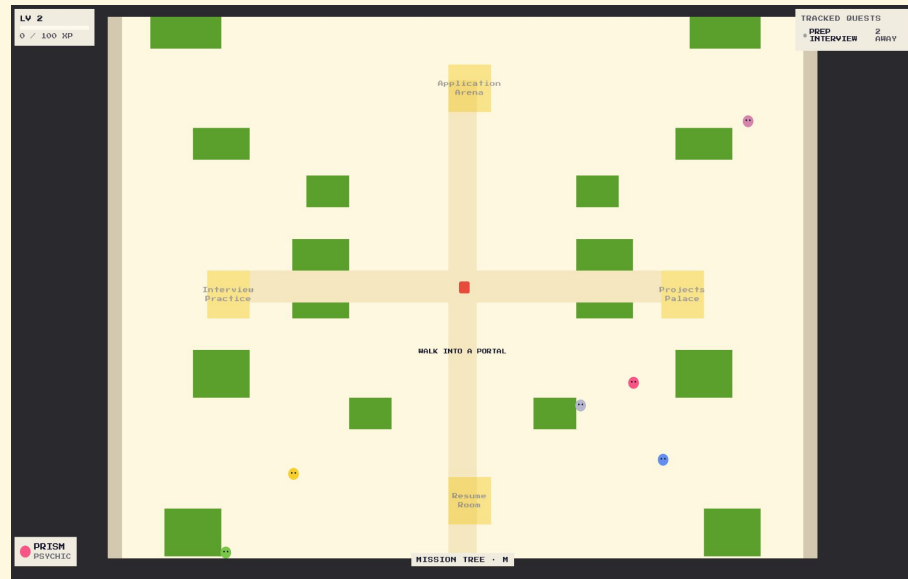
Users expressed confusion on how to catch creatures, and what objects were and were not creatures. We updated the creature design with a face and a floating instruction for how to catch the creature.

Spawn Location

Before

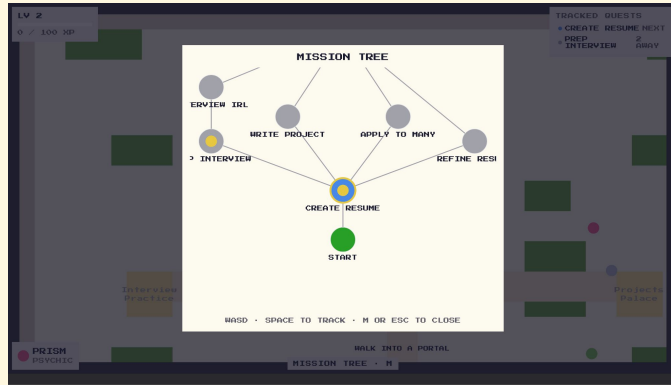


After

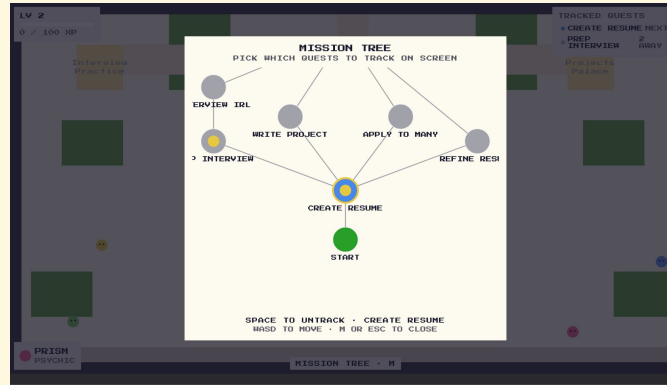


Users were frustrated by the player spawn location right in front of the resume room; any forward movement would transport the player there, even when that wasn't the goal. We moved the spawn to the center. This also gave users a full view of the entire game.

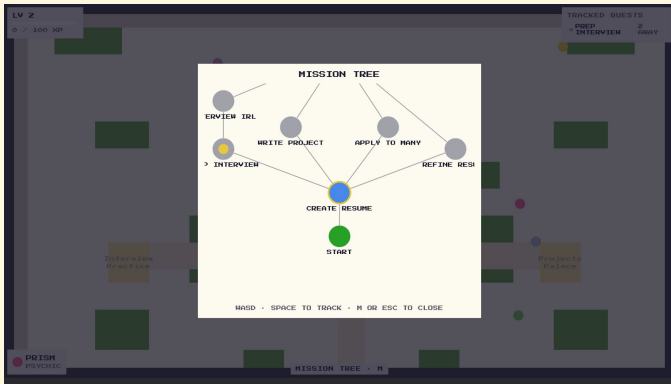
Mission Tree



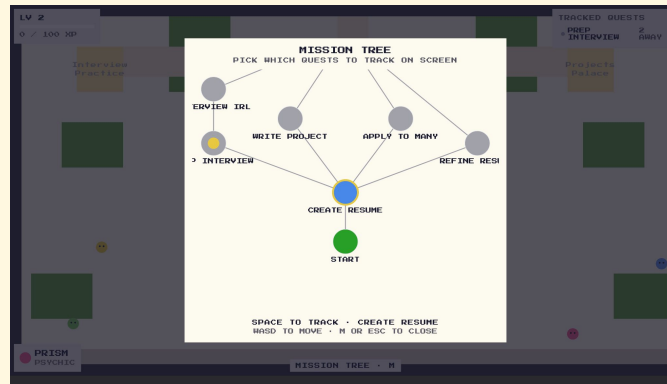
Selected - Before



Selected - After



Unselected - Before



Unselected - After

The mission tree could be confusing: users were unable to identify how to pin & track task.

We highlighted important instructions, making it easier to read.

We also added a general introduction instruction, "Pick which quests to track on screen" so users have a clear goal when using the mission tree.

Tutorials

Company	Role	Posted	Applicants
Meta	Backend SWE — Analytics & Recommendations	00:57:06	12 apps
Stripe	Infrastructure Engineer — Payments Core	01:14:25	8 apps
Figma	Frontend Engineer — Canvas Infrastructure	02:13:13	34 apps
Microsoft	Data Engineer — Azure Analytics Platform	14:56:00	452 apps

Company	Role	Status
Airbnb	Platform Engineer	PENDING
Coinbase	Backend SWE	REJECTED
Stripe	SWE Infrastructure	QA RECEIVED
Figma	Frontend Engineer	INTERVIEW

Before

Recent Listings

This page shows listings tailored to your resume and history. Click a role to review its details and continue the apply flow.

Close Next

After (First Visit)

MARKET INTELLIGENCE

0.05% Predicted odds

0.1% Accept rate

Aggregated from Reddit · Glassdoor · Levels.fyi · WSO

COMMUNITY INTEL

REDDIT COMPANY SITE Two Sigma campus hire: matched to team 2 months before start. My team was distributed r/cscareerquestions · 388 upvotes · 6 days ago

GLASSDOOR COMPANY SITE Collaborative and intellectually honest culture. Engineering very much appreciated. Glassdoor · ★★★★★ · 2 weeks ago

LEVELS.FYI COMPANY SITE Campus hire SWE: \$165k-\$250k base + performance bonus up to 30%. Onsite gym, meals, strong benefits. Levels.fyi · 12 verified datapoints

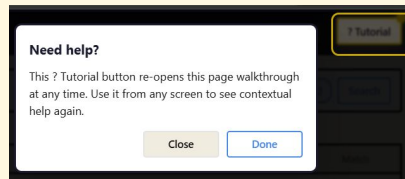
Before

Community Intel

These community insights are scraped from public candidate reports and discussion forums. They help you understand company-specific interview patterns and application preferences.

Close Next

After (First Visit)



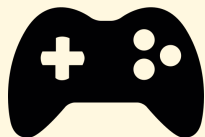
The Application Arena could be confusing: users were unsure what features were available and how to use them effectively.

We highlighted important information through guided tutorial pop-ups, making key tools easier to understand.

We also added first-visit walkthroughs so users have a clearer understanding of job listings, application tracking, and community insights when exploring the page.

LIMITATIONS

User Assumptions



Users are somewhat familiar with 2D RPG game conventions

Job Quest uses mechanics such as exploration, quest tracking, NPC interactions, and world navigation that are common in RPG games. Users with little familiarity may require additional guidance to understand how to progress through the game.



Users are motivated by gamification

The prototype assumes that features like quests, XP, creature encounters, and progression systems will increase engagement and motivation. However, some users may prefer a more traditional career-preparation experience and may not find game elements motivating.



Users are actively searching for jobs or internships

Many of the game's tools and activities are designed around the needs of students currently navigating the job-search process. Users who are not actively seeking opportunities may find fewer features relevant to their immediate goals.

Unaddressed User Needs

Personalized feedback on applications and interviews

While Job Quest encourages progress and practice, it does not currently provide detailed, individualized feedback on resumes, cover letters, or interview responses. Several users expressed interest in more actionable guidance.

Realistic career pathway recommendations

Our domain expert emphasized that many students have unrealistic expectations about entry-level positions. Although the game promotes career preparation, it does not currently provide personalized recomm

Future Personas

Non-Tech Job Seekers

Applications have similar structure and remain easy targets for automation, while conserving personal touch. Non-tech roles hire more through in-person and networking channels, so the application arena should support interaction volume instead.

Users still benefit from synthetic motivation and feedback from gamification: mission tree is curated to hold domain-specific quests, and creatures retain industry-specific knowledge prompts.

The same also holds for other application types, such as internships, grants, and research.

International Students

A new distinct pains current personas don't face: visa status viability and lottery deadlines add time pressure.

The engine remains unchanged: game loops target the same consistency pain that motivated gamification. Arena extensions include visa-status filter on listings, open-to-sponsorship employer tags, and ATS sponsorship-history checks. Mission tree can include a new visa track sub-tree for paperwork milestones.

This problem is especially relevant currently due to the politically-imposed difficulty of obtaining visas.

REFERENCES

References

- Hooft, E. A. J. van , Mueller, J. D. K., Wanberg, C. R., Kanfer, R., & Basbug, G. (2020). Job search and employment success: A quantitative review and future research agenda. *Journal of Applied Psychology*, 106(5).
<https://doi.org/10.1037/apl0000675>
- Liu, S., Huang, J. L., & Wang, M. (2014). Effectiveness of job search interventions: a meta-analytic review. *Psychological bulletin*, 140(4), 1009–1041. <https://doi.org/10.1037/a0035923>
- da Motta Veiga, S. P., & Turban, D. B. (2018). Insight into job search self-regulation: Effects of employment self-efficacy and perceived progress on job search intensity. *Journal of Vocational Behavior*, 108, 57–66.
<https://doi.org/10.1016/j.jvb.2018.06.010>